

Skill Arena - Phase 6 Document 3

Analytics Platform, Business Intelligence, Data Warehouse & Executive Insights Architecture

Document Purpose

Define the data analytics architecture used to measure platform performance, player behaviour, financial health and strategic growth.

Business Objective

Transform operational data into actionable intelligence for product, finance, marketing and executive teams.

Analytics Vision

Every major business decision should be supported by measurable platform intelligence.

Core Components

Analytics Platform, Data Warehouse, Business Intelligence Layer, Reporting Engine and Executive Insights Engine.

Data Collection Framework

Collect gameplay, financial, operational, tournament, security and engagement data.

Event Collection

Centralized event ingestion from all platform services.

Data Warehouse

Central repository optimized for reporting, forecasting and business analysis.

Analytics Domains

Player Analytics, Gameplay Analytics, Financial Analytics, Tournament Analytics and Security Analytics.

Player Analytics

Track retention, progression, engagement, churn risk and player lifecycle metrics.

Gameplay Analytics

Measure challenge completion, difficulty performance, matchmaking effectiveness and replay activity.

Financial Analytics

Monitor deposits, withdrawals, treasury performance, liabilities and revenue.

Tournament Analytics

Track participation, qualification rates, engagement and championship performance.

Security Analytics

Analyze fraud trends, account risks, incident rates and enforcement outcomes.

Business Intelligence Layer

Provides dashboards, reports and self-service analysis tools.

Executive Dashboards

Growth metrics, treasury health, revenue, retention and operational performance.

Operational Dashboards

Service health, platform activity, tournament activity and user engagement.

Forecasting Engine

Project growth, treasury requirements, reward liabilities and infrastructure demand.

Data Governance

Ensure accuracy, consistency, ownership and lifecycle management of platform data.

Data Quality Controls

Validation, cleansing, monitoring and anomaly detection.

Real-Time Analytics

Support live dashboards and operational decision making.

Historical Analytics

Long-term trend analysis and strategic planning support.

Scalability Requirements

Support billions of events and long-term historical analysis.

Security Requirements

Analytics data protected through role-based access and data governance controls.

Audit Requirements

Reporting access, dashboard activity and data changes permanently logged.

Future Expansion

Foundation for AI analytics, predictive modelling and advanced business intelligence.

Acceptance Criteria

Platform leadership can measure, analyze and optimize all major business functions.